ENGINEERING CHEMISTRY

LEC - 1

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SCHRÖDINGER WAVE EQUATION

IMPORTANT THEORY QUESTIONS

1. Who proposed the Schrödinger wave equation?

The Schrödinger wave equation was proposed by Erwin Schrödinger in 1926.

2. What is a wave function (ψ)?

The wave function, denoted by ψ (psi), is a mathematical function that describes the Quantum state of a particle. It contains all the information about the system and its behaviour.

3. State the de Broglie equation.

According to de Broglie hypothesis, every moving particle is associated with a wave. The equation is:

$$\lambda = \frac{h}{mv}$$

where, λ = wavelength of the particle
h = Planck's constant
m = mass of the particle
v = velocity of the particle

4. What does ψ^2 represent?

ψ^2 (psi squared) represents the probability density of finding the particle at a particular point in space. It gives the probability of locating the electron within a small volume element.

5. Write the time-independent Schrödinger wave equation.

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0$$

where,
ψ = wave function
m = mass of electron
h = Planck's constant
E = total energy
V = potential energy

Note:
- E = Total energy of the electron
- V = Potential energy experienced by the electron
- $(E-V)$ = Kinetic energy of the electron

Key Point:
This equation helps us to understand the wave nature of electrons and predict their behaviour in atoms and molecules.

SCHRÖDINGER WAVE EQUATION

IMPORTANT THEORY QUESTIONS

6. **What is the physical meaning of V in the Schrödinger equation?**

In the Schrödinger equation, V represents the potential energy. It is the potential energy experienced by the electron due to the nucleus and other electrons in the atom or molecule. It varies from point to point in space.

7. **What does E represent in the Schrödinger equation?**

In the Schrödinger equation, E represents the total energy of the electron in the system. It is the sum of kinetic energy and potential energy of the electron.

$$E = K.E. + P.E.$$

Note:
E is a constant for a given stationary state.

8. **State Heisenberg's uncertainty principle.**

Heisenberg's uncertainty principle states that it is impossible to determine simultaneously and exactly both the position and momentum of a particle.

Mathematically, $$\Delta x \cdot \Delta p_x \geq \frac{h}{4\pi}$$

where,
Δx = uncertainty in position
Δp_x = uncertainty in momentum
h = Planck's constant

9. **What is the significance of the wave function?**

The wave function (ψ) completely describes the quantum state of a system. It contains all the information about the system and allows us to calculate the probability of finding the particle in any region of space.

10. **Write any one importance of the Schrödinger equation.**

The Schrödinger equation is very important because it helps us to determine the energy levels and wave functions of electrons in atoms and molecules, and thus predicts their behaviour.

Importance:
★ It explains the wave nature of electrons and helps in predicting atomic structure and spectra.

Key Point:
★ ψ → wave function (mathematical description)
★ ψ^2 → probability density of finding the electron
★ E → total energy (constant for a stationary state)
★ V → potential energy

Remember:
More ψ^2 at a point means higher chance of finding the electron there.

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SCHRÖDINGER WAVE EQUATION

IMPORTANT THEORY QUESTIONS

1. Explain the concept of the wave function (ψ) and its significance.

The wave function, ψ (psi), is a mathematical function that describes the quantum state of a particle. It depends on the spatial coordinates (x, y, z) and time (t), i.e., $\psi = \psi(x, y, z, t)$.

Significance :

- $|\psi|^2$ gives the probability density of finding the particle at a particular point in space.
- ψ contains all observable information about the system.
- It must be single valued, finite, continuous, and square integrable.
- Physically meaningful solutions of ψ are called acceptable wave functions.

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2. State and explain the Schrödinger wave equation.

The Schrödinger wave equation describes the behavior of a quantum system. It is a fundamental equation of quantum mechanics.

Time-dependent form :

$$-\frac{h^2}{8\pi^2 m}\nabla^2\psi + V\psi = ih\frac{\partial\psi}{\partial t}$$

This equation is a second-order linear partial differential equation. It gives the relationship between the wave function ψ and the potential energy V of the system.

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3. Explain the physical meaning of each term in the Schrödinger equation.

$$-\frac{h^2}{8\pi^2 m}\nabla^2\psi + V\psi = ih\frac{\partial\psi}{\partial t}$$

- $-\frac{h^2}{8\pi^2 m}\nabla^2\psi$: Kinetic energy operator. It represents the kinetic energy associated with the particle.
- $V\psi$: Potential energy term. Here, V is the potential energy of the particle in the system.
- $ih\frac{\partial\psi}{\partial t}$: Time dependent term. It represents the change in wave function with time.
- ψ : Wave function. It contains complete information about the system.
- h : Planck's constant (6.626×10^{-34} J·s).
- m : Mass of the particle.
- ∇^2 : Laplacian operator. In Cartesian coordinates, $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

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4. Write the assumptions behind Schrödinger wave mechanics.

- The particle is an associated wave.
- The wave function ψ is continuous, single valued and finite everywhere.
- ψ is square integrable, i.e., $\int |\psi|^2 \, d\tau = 1$ (normalization condition).
- The total energy of the system is conserved.
- The potential energy V is a real function.
- The classical concepts are not valid at atomic and subatomic scale.

SCHRÖDINGER WAVE EQUATION

IMPORTANT THEORY QUESTIONS

5. Explain the importance/applications of the Schrödinger equation.

The Schrödinger equation is one of the most important equations in quantum mechanics. Its major applications are:

- It gives a complete description of the quantum state of a system.
- It is used to calculate energy levels and wave functions of atoms and molecules.
- It helps in predicting the shapes of atomic orbitals (s, p, d, f) and molecular orbitals.
- It explains chemical bonding, molecular structure and stability.
- It is the basis for modern spectroscopic techniques like UV-Vis, IR, NMR, etc.
- It is widely used in the fields of solid-state physics, nanotechnology, quantum chemistry and material science.
- It allows the calculation of probability densities, electron distributions and transition probabilities.

6. Describe the relation between de Broglie hypothesis and Schrödinger equation.

The de Broglie hypothesis states that matter particles exhibit wave nature.

According to de Broglie,

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

where, h = Planck's constant, m = mass of particle, v = velocity.

Schrödinger used this wave nature of particles and developed a wave equation to describe the behaviour of electrons in atoms. Thus, de Broglie provided the conceptual basis (matter waves), while Schrödinger gave the mathematical formulation (wave equation) to describe them.

Mathematically, Ψ (wave function) represents the matter wave associated with the particle as proposed by de Broglie.

7. Explain why Schrödinger's model is superior to Bohr's model.

Schrödinger's model is considered superior to Bohr's model because:

1. It is based on quantum mechanics and is more fundamental.
2. It applies to multi-electron atoms and molecules, whereas Bohr's model is valid only for hydrogen-like atoms.
3. It explains the shapes of orbitals (s, p, d, f) and their orientations.
4. It accounts for spectral fine structure and relative intensities of spectral lines.
5. It is consistent with Heisenberg's uncertainty principle.
6. It provides a probability interpretation ($|\Psi|^2$) of electron presence, not a fixed path.
7. It successfully explains bond formation, molecular spectra and chemical reactivity.
8. It forms the basis of modern chemistry and physics.

Bohr's Model	Schrödinger's Model
Valid only for hydrogen atom	Valid for all atoms and molecules
Based on fixed circular orbits	Based on probability waves (orbitals)
Cannot explain fine structure	Explains fine structure and intensities
No concept of orbital shape	Explains shapes and orientation of orbitals
Semi-classical approach	Fully quantum mechanical approach

SCHRÖDINGER WAVE EQUATION

IMPORTANT THEORY QUESTION (5/15 MARKS)

1. Derive the time-independent Schrödinger wave equation.

Answer: We start from the time-dependent Schrödinger wave equation and separate the time and space variables to obtain the time-independent equation.

Step 1 : Time-dependent Schrödinger wave equation

The fundamental equation is,

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\vec{x},t) + V(\vec{x},t)\psi(\vec{x},t) = i\hbar\frac{\partial\psi(\vec{x},t)}{\partial t} \quad \ldots(1)$$

where, $\psi(\vec{x},t)$ = wave function
$V(\vec{x},t)$ = potential energy
∇^2 = Laplacian operator.

Step 2 : Assumption of separability

For a system where the potential energy does not depend on time ($V = V(\vec{x})$ only), we assume that the wave function can be written as

$$\psi(\vec{x},t) = \psi(\vec{x})\,T(t) \quad \ldots(2)$$

where, $\psi(\vec{x})$ = spatial part (depends only on $\vec{x}$)
$T(t)$ = time part (depends only on t)

Step 3 : Substitute in equation (1)

Substituting (2) in (1), we get

$$-\frac{\hbar^2}{2m}T(t)\nabla^2\psi(\vec{x}) + V(\vec{x})\psi(\vec{x})T(t) = i\hbar\,\psi(\vec{x})\frac{dT(t)}{dt}$$

Step 4 : Separate the variables

Divide both sides by $\psi(\vec{x})\,T(t)$,

$$-\frac{\hbar^2}{2m}\frac{\nabla^2\psi(\vec{x})}{\psi(\vec{x})} + V(\vec{x}) = i\hbar\frac{1}{T(t)}\frac{dT(t)}{dt}$$

Left side depends only on $\vec{x}$ and right side depends only on t.
Therefore, both sides must be equal to a constant (separation constant).
Let this constant be E (which will represent the total energy of the system).

Step 5 : Time part

$$i\hbar\frac{1}{T(t)}\frac{dT(t)}{dt} = E$$

$$\frac{dT(t)}{dt} = -\frac{iE}{\hbar}T(t)$$

Integrating,

$$T(t) = Ae^{-iEt/\hbar}$$

where, A = constant

Step 6 : Space part (Time-independent Schrödinger equation)

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\vec{x}) + V(\vec{x})\psi(\vec{x}) = E\psi(\vec{x})$$

This is the time-independent Schrödinger wave equation.
It gives the spatial distribution of the particle in a stationary state with energy E.

Summary :

The time-dependent wave function is,

$$\psi(\vec{x},t) = \psi(\vec{x})\,e^{-iEt/\hbar}$$

and the time-independent Schrödinger equation is,

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\vec{x}) + V(\vec{x})\psi(\vec{x}) = E\psi(\vec{x})$$

Where,
$\hbar$ = Planck's constant $= 1.054 \times 10^{-34}$ J·s
m = mass of the particle
$V(\vec{x})$ = potential energy
E = total energy

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SCHRÖDINGER WAVE EQUATION

IMPORTANT THEORY QUESTION (5/15 MARKS)

2. Explain the Schrödinger wave equation with its assumptions and significance.

Schrödinger Wave Equation :

The Schrödinger wave equation is a fundamental equation of quantum mechanics which describes the behaviour of a quantum system in terms of its wave function.

It is of two types : (i) Time-dependent equation (ii) Time-independent equation

- **Time-dependent Schrödinger wave equation :**

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\vec{r},t) + V(\vec{r},t)\psi(\vec{r},t) = i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t}$$

where,
$\psi(\vec{r},t)$ = wave function
$V(\vec{r},t)$ = potential energy
m = mass of particle
$\hbar$ = $\frac{h}{2\pi}$ (Planck's constant)
∇^2 = Laplacian operator
i = $\sqrt{-1}$

- **Time-independent Schrödinger wave equation :**

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\vec{r}) + V(\vec{r})\psi(\vec{r}) = E\psi(\vec{r})$$

where,
$\psi(\vec{r})$ = spatial wave function
E = total energy of the system

Assumptions :

1. The particle is associated with a wave described by the wave function ψ.
2. The wave function ψ is single-valued, finite and continuous everywhere.
3. ψ is square integrable, i.e., $\int |\psi|^2 d\tau = 1$ (normalization condition).
4. The total energy of the system is conserved.
5. The potential energy V is a real function.
6. The classical concepts of definite path and position are not valid at atomic scale.

Significance :

1. It provides a mathematical description of quantum systems.
2. It gives the probability density as $|\psi|^2$, which represents the likelihood of finding the particle at a particular point.
3. It allows calculation of energy levels and wave functions of atoms and molecules.
4. It explains the shapes of atomic and molecular orbitals.
5. It forms the basis for all advanced quantum mechanical theories.
6. It is applicable to systems with any potential energy function.
7. It successfully accounts for phenomena like tunneling effect, interference, zero-point energy, etc.
8. It is widely used in spectroscopy, chemical bonding, solid state physics, nanotechnology and modern chemistry.

SCHRÖDINGER WAVE EQUATION

IMPORTANT THEORY QUESTION (5/15 MARKS)

3. Discuss the importance and applications of the Schrödinger wave equation.

A. IMPORTANCE

The Schrödinger wave equation is one of the most important equations in quantum mechanics. Its importance can be summarized as follows :

① It is a complete and fundamental equation of quantum mechanics.
② It describes the behaviour of microscopic particles (electrons, protons, atoms, molecules, etc.) in terms of their wave nature.
③ It provides exact mathematical treatment of quantum systems, unlike semi-classical models.
④ It allows calculation of wave function (ψ) and therefore probability density ($|\psi|^2$).
⑤ It gives allowed energy levels and eigenfunctions (orbitals) of the system.
⑥ It explains stability of atoms and chemical bonding in molecules.
⑦ It is consistent with all experimental observations at microscopic level.
⑧ It forms the basis for all advanced theories such as quantum chemistry, solid state physics, spectroscopy, etc.

B. APPLICATIONS

The Schrödinger wave equation has wide range of applications in various fields of science and technology.

① Atomic Structure :
- Solving the Schrödinger equation for the hydrogen atom gives exact energy levels and wave functions (orbitals).

$$E_n = -\frac{13.6}{n^2}\ \text{eV} \quad (n = 1, 2, 3, \ldots)$$

② Molecular Structure and Bonding :
- Explains the shape of molecules, bond lengths, bond energies and molecular orbitals.
- Basis of Valence Bond Theory and Molecular Orbital Theory.

③ Spectroscopy :
- Used to interpret electronic, vibrational and rotational spectra of atoms and molecules.

Selection rule :

$$\Delta l = \pm 1$$
$$\Delta m = 0, \pm 1$$

④ Chemical Reactivity :
- Wave function helps in understanding reactivity, transition states and reaction mechanisms.

⑤ Solid State Physics :
- Describes the behaviour of electrons in crystals, band structure, conductivity and other electrical properties.

Particle in a box model :

$$E_n = \frac{n^2 h^2}{8mL^2}$$
$$(n = 1, 2, 3, \ldots)$$

⑥ Quantum Chemistry :
- Basis for ab initio calculations, Hartree-Fock method, DFT, etc. to predict molecular properties.

⑦ Nanotechnology :
- Helps in designing and understanding quantum dots, nanowires and nanoscale devices.

⑧ Nuclear Physics :
- Used to study the behaviour of nucleons in nucleus (nuclear shell model).

⑨ Quantum Tunneling :
- Explains tunneling effect of particles through potential barriers, important in devices like tunnel diode, scanning tunneling microscope (STM), etc.

Tunneling probability

$$P \approx e^{-\frac{2}{\hbar}\int_{x_1}^{x_2}\sqrt{2m(V-E)}\,dx}$$

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SCHRÖDINGER WAVE EQUATION

IMPORTANT THEORY QUESTION (5/15 MARKS)

4. Explain the derivation of the Schrödinger equation from de Broglie's hypothesis.

A. de Broglie's Hypothesis :

de Broglie postulated that every moving particle is associated with a wave whose wavelength (λ) is given by

$$\lambda = \frac{h}{p}$$

(where, $\hbar$ = Planck's constant, p = momentum of the particle)

B. Derivation of Schrödinger Wave Equation from de Broglie's Hypothesis :

Consider a free particle of mass m moving along the x-axis.
According to de Broglie, the associated wave is a matter wave.

Plan of derivation :
(i) Start from wave equation
(ii) Use de Broglie relation
(iii) Obtain energy operator
(iv) Substitute to get the Schrödinger equation

Step 1 : Start from classical wave equation

The general one-dimensional wave equation is

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 \psi}{\partial t^2} \quad \ldots (1)$$

Where,
ψ = wave function
v = velocity of the wave

Step 2 : Relate wave velocity with particle velocity

The velocity of the wave is $v = \frac{\omega}{k}$ Where, ω = angular frequency, k = wave number

For a particle, momentum $p = \hbar k$ (since $\lambda = \frac{h}{p} \Rightarrow k = \frac{2\pi}{\lambda} \Rightarrow \hbar k = p$)

Also, energy $E = \hbar\omega$

$$\hbar = \frac{h}{2\pi}$$

Reduced Planck's constant

Step 3 : Express time derivative in terms of energy

From $E = \hbar\omega \Rightarrow \omega = \frac{E}{\hbar}$

Also, $p = \hbar k \Rightarrow k = \frac{p}{\hbar}$

Step 4 : Substitute in wave equation

Substitute ω and k in equation (1)

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{(\omega/k)^2}\frac{\partial^2 \psi}{\partial t^2} = \frac{k^2}{\omega^2}\frac{\partial^2 \psi}{\partial t^2}$$

$$= \frac{\frac{p^2}{\hbar^2}}{\frac{E^2}{\hbar^2}}\frac{\partial^2 \psi}{\partial t^2} = \frac{p^2}{E^2}\frac{\partial^2 \psi}{\partial t^2}$$

Step 5 : Rearrange and introduce operators

Rearranging,

$$p^2 \psi = -E^2 \frac{\partial^2 \psi}{\partial t^2} \times \left(-\frac{\hbar^2}{p^2}\right)$$

$$\Rightarrow -\hbar^2 \frac{\partial^2 \psi}{\partial x^2} = -E^2 \frac{\partial^2 \psi}{\partial t^2}$$

Now, use operator form,

$$\hat{p} = -i\hbar\frac{\partial}{\partial x}, \quad \hat{E} = i\hbar\frac{\partial}{\partial t}$$

Substituting and simplifying,

$$i\hbar\frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} \quad \ldots (2)$$

This is the **time-dependent Schrödinger equation**

Step 6 : Include potential energy

For a particle in potential field $V(x)$, total energy operator becomes

$$\hat{E} = i\hbar\frac{\partial}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)$$

Hence, the time-dependent Schrödinger equation is

$$i\hbar\frac{\partial \psi}{\partial t} = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right]\psi \quad \ldots (3)$$

If the potential is independent of time, we can separate variables and obtain the **time-independent Schrödinger equation**.

$$-\frac{\hbar^2}{2m}\frac{d^2 \psi}{dx^2} + V(x)\psi = E\psi \quad \ldots (4)$$

This is the **time-independent Schrödinger equation**.

Summary of Derivation :

Starting from de Broglie's hypothesis ($\lambda = \frac{h}{p}$), the wave nature of a particle is used in the classical wave equation, and by relating frequency and wave number with energy and momentum, we obtain the Schrödinger wave equation in operator form.

Key Relations Used :

- $\lambda = \frac{h}{p}$
- $p = \hbar k$, $E = \hbar\omega$
- $\hat{p} = -i\hbar\frac{\partial}{\partial x}$, $\hat{E} = i\hbar\frac{\partial}{\partial t}$
- $\hbar = \frac{h}{2\pi}$

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SCHRÖDINGER WAVE EQUATION

IMPORTANT THEORY QUESTION (5/15 MARKS)

5. Write short notes on:

(a) Wave function (ψ) (b) Probability density (c) Importance of Schrödinger equation

(a) WAVE FUNCTION (ψ) :

- In quantum mechanics, the wave function denoted by ψ (psi) is a mathematical function which **completely describes the quantum state** of a particle.
- It depends on the spatial coordinates (x, y, z) and time (t).
- For a stationary state, it depends only on spatial coordinates : $\boxed{\psi = \psi(x, y, z)}$
- It is in general a complex function and can be written as,

$$\boxed{\psi(x, y, z, t) = \psi_r(x, y, z, t) + i\,\psi_i(x, y, z, t)} \quad [\psi_r = \text{real part}, \ \psi_i = \text{imaginary part}]$$

- ψ itself has no direct physical meaning but $|\psi|^2$ (its modulus square) **has physical significance**.
- The wave function must satisfy the following conditions :
  1. ψ must be single valued, finite and continuous everywhere.
  2. ψ must be square integrable : $\boxed{\int_{-\infty}^{+\infty} |\psi|^2 \, d\tau = 1}$ (Normalization condition)
  3. ψ must be zero at infinity (i.e., $\psi \to 0$ as $x \to \infty$).
  4. $\frac{d\psi}{dx}$, $\frac{d\psi}{dy}$, $\frac{d\psi}{dz}$ must be finite and continuous.

$d\tau$ = volume element = $dx\,dy\,dz$ (in Cartesian coordinates)

(b) PROBABILITY DENSITY :

- The square of the modulus of the wave function, $|\psi|^2$, is called the **probability density** or **probability distribution function**.
- $|\psi(x, y, z, t)|^2 \, d\tau$ gives the probability of finding the particle in the volume element $d\tau$ at the point (x, y, z) at time t.
- Thus, $\boxed{\text{Probability density} = |\psi(x, y, z, t)|^2}$
- It has the following important properties :
  - ★ It is always real and positive.
  - ★ Its integral over the entire space is equal to 1. $\boxed{\int_{-\infty}^{+\infty} |\psi|^2 \, d\tau = 1}$
  - ★ A larger value of $|\psi|^2$ at a point means higher probability of finding the particle there.

(c) IMPORTANCE OF SCHRÖDINGER EQUATION :

- ✿ It is the **fundamental equation** of quantum mechanics and provides a **complete mathematical description** of quantum systems.
- ✿ It allows calculation of **wave functions** and **energy values** (eigenvalues) of atoms and molecules.
- ✿ It explains the **shape** and behavior of **atomic and molecular orbitals**.
- ✿ It is applicable to **any potential energy function**, unlike Bohr's model which is restricted to hydrogen-like atoms.
- ✿ It forms the basis for **advanced theories** and **modern chemistry**, solid state physics, spectroscopy, nanotechnology, etc.
- ✿ It successfully explains phenomena such as **tunneling effect**, **interference**, **zero-point energy**, etc.

For example :

- ★ Atomic spectra
- ★ Chemical bonding
- ★ Molecular structure
- ★ Reaction mechanism
- ★ Band theory of solids
- ★ Quantum dots
- ★ Nanodevices
- ★ Laser and semiconductors

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PYQ

Q. Prove that $\left[\nabla^2 + \frac{8\pi^2 m}{h^2}(E - V)\right]\psi = 0$ [2022-2023]

Ans.

Starting from the time-independent Schrödinger wave equation,

$$-\frac{h^2}{8\pi^2 m}\nabla^2\psi + V\psi = E\psi \quad \ldots(1)$$

Rearranging equation (1),

$$-\frac{h^2}{8\pi^2 m}\nabla^2\psi = E\psi - V\psi$$

$$-\frac{h^2}{8\pi^2 m}\nabla^2\psi = (E - V)\psi$$

Multiplying both sides by $-\frac{8\pi^2 m}{h^2}$,

$$\nabla^2\psi = -\frac{8\pi^2 m}{h^2}(E - V)\psi$$

Rearranging,

$$\left[\nabla^2 + \frac{8\pi^2 m}{h^2}(E - V)\right]\psi = 0$$

Hence proved. ★

Where,

- $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ (Laplacian operator)
- ψ = Wave function
- V = Potential energy
- E = Total energy
- m = Mass of the particle
- h = Planck's constant

Key Points :

- ★ The equation is known as Helmholtz equation.
- ★ It relates the wave function ψ with the potential energy V and total energy E.
- ★ It is a fundamental equation in quantum mechanics.

Note : This equation is valid only for time-independent systems where E is a constant.

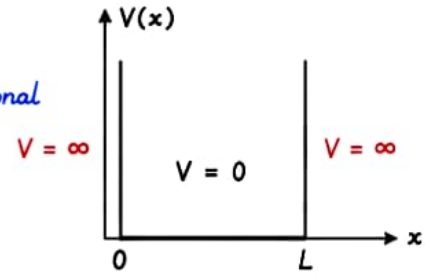
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PYQ

Q. Set up the Schrödinger equation for a particle in a one-dimensional box. [2023-2024]

Ans.

Consider a particle of mass m confined in a one-dimensional box extending from $x = 0$ to $x = L$.



One-dimensional box

1. Potential energy function, $V(x)$:

$$V(x) = \begin{cases} 0 & \text{for } 0 < x < L \quad \text{(inside the box)} \\ \infty & \text{for } x \le 0 \text{ or } x \ge L \quad \text{(outside the box)} \end{cases}$$

2. Time-independent Schrödinger equation :

$$-\frac{h^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

$\psi(x)$ = wave function
E = energy eigenvalue
m = mass of particle
$\hbar$ = Planck's constant / 2π

3. Inside the box ($0 < x < L$) : $V(x) = 0$

The equation becomes,

$$-\frac{h^2}{2m}\frac{d^2\psi(x)}{dx^2} = E\psi(x)$$

or, $\frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0$, where $k^2 = \frac{2mE}{h^2}$

General solution : $\psi(x) = A \sin kx + B \cos kx$

4. Boundary conditions :

- At $x = 0$ (infinite potential wall) $\Rightarrow \psi(0) = 0$
  $\Rightarrow B \cos 0 = B = 0 \Rightarrow B = 0$
- At $x = L$ (infinite potential wall) $\Rightarrow \psi(L) = 0$
  $\Rightarrow A \sin kL = 0 \Rightarrow \sin kL = 0$
  $\Rightarrow kL = n\pi$; $n = 1, 2, 3, \ldots$

Key points :

- ★ $\psi(x)$ must be finite and single valued everywhere.
- ★ $\psi(x) = 0$ at the walls of the box.
- ★ Only discrete energy values are allowed.

5. Allowed values and wave functions :

- $k_n = \frac{n\pi}{L}$; $n = 1, 2, 3, \ldots$
- $\psi_n(x) = A \sin \frac{n\pi x}{L}$, $0 < x < L$ (normalized wave functions)
- Energy eigenvalues :

$$E_n = \frac{h^2 k_n^2}{2m} = \frac{n^2\pi^2 h^2}{2mL^2}, \quad n = 1, 2, 3, \ldots$$

[Energy levels are discrete and increase as n^2]

Note : This is the fundamental model for understanding quantization of energy.